Swami Keshvanand Institute of Technology,Management & Gramothan, Jaipur

Department of Computer Science & Engineering

Final Project Evaluation Report

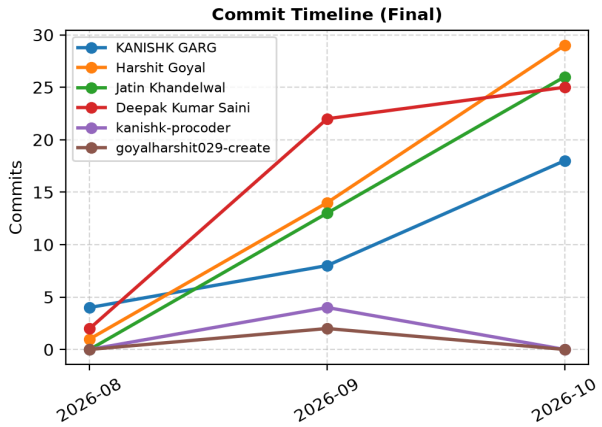
Project Repository: SKIT-AI-2023-2027-14 | **Branch:** main

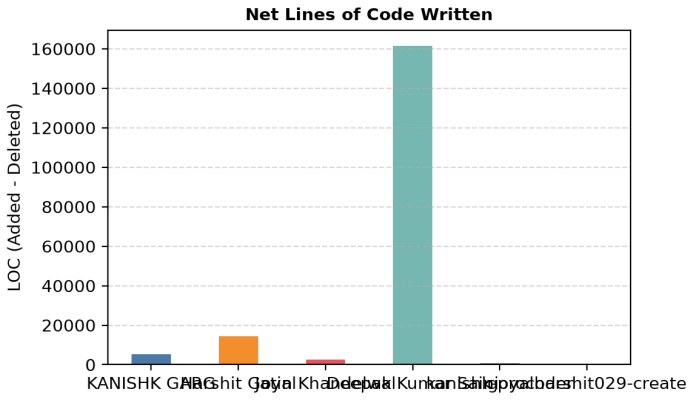
Evaluation Window: Complete Project Lifecycle (All Commits) | **Generated On:** October 30, 2026

1. Individual Contribution Breakdown

Student Name	Commits (%)	Lines Added	Lines Deleted	Net LOC	Active Days
KANISHK GARG	30 (17.9%)	+17,800	-12,531	5,269	10 days
Harshit Goyal	44 (26.2%)	+183,545	-169,097	14,448	12 days
Jatin Khandelwal	39 (23.2%)	+2,823	-281	2,542	11 days
Deepak Kumar Saini	49 (29.2%)	+165,385	-3,992	161,393	13 days
kanishk-procoder	4 (2.4%)	+761	-1	760	2 days
goyalharshit029-create	2 (1.2%)	+423	-0	423	1 days

2. Visual Trends & Volume





3. Detailed Commit Logs & Mentor Evaluation (Final)

Student: KANISHK GARG — 30 commit(s)

Date	Hash	Commit Message	Mentor Marks (/10)
2026-10-30	80f8776	Refactor generate_report.py for improved readability	_____ / 10
2026-10-30	ffbfb23	Add workflow to generate and commit project report	
2026-10-30	f4fceec	final report generated	
2026-10-30	74c929c	Add script to generate consolidated project report	
2026-10-30	55c31e9	git action workflow : summary report	
2026-10-20	b0a05e1	developed api for ecg prediction model	
2026-10-18	cec70bc	pdf report - updation in format of this	
2026-10-18	293fde3	solved error of printed report generation	
2026-10-18	e84259f	added the report print feature to the prediction page	
2026-10-18	a600e5c	the prediction api for the clinical model is connected to a separate result page and generating report	
2026-10-16	a4cd69b	ADDED DIRECT NEVIGATIONS	

Date	Hash	Commit Message	Mentor Marks (/10)
2026-10-15	3cc8076	prediction api connected	
2026-10-09	c473d57	updated upload api and retreive previous uploads too	
2026-10-08	18f11f2	resolved error in data upload api connection with frontend backend and database	
2026-10-08	b172627	dashboard connection	
2026-10-08	2a0e55b	made some changes in backend and frontend so they will work together finely	
2026-10-08	d87d172	Update .env.example	
2026-10-08	064a809	Create CVD-XAI_Weekly_Progress_Report_Form-3_2026-09-25.pdf	
2026-09-25	af55d37	feat(api): pass db session in patient lookup and add prediction endpoint tests	
2026-09-25	a664323	TESTED PATIENT API	
2026-09-25	0b4db49	UPDATED PATIENT API TO GET CORRECT PREDICTION	
2026-09-25	7b95c97	Update ml_engine.py	
2026-09-25	8f6c0cc	MODEL SELECTION FOR THE CLINIC DATA	
2026-09-10	940f6c9	omplete Sprint 2(A) — database health checks, verification, and seed data	
2026-09-10	a2c2b34	complete Sprint 1 — FastAPI server, JWT auth, and data upload APIs	
2026-09-10	bc2c2d6	refactor: clean up backend — remove seed scripts, fix main.py for MongoDB	
2026-08-10	869059d	small deletion in backend data	
2026-08-10	b1f22de	small deletion in backend data	
2026-08-10	a52e6c0	backend word day 1	
2026-08-10	f38d84a	backend word day 1	

Student: Harshit Goyal — 44 commit(s)

Date	Hash	Commit Message	Mentor Marks (/10)
2026-10-29	e36744b	chore: update patient registration notes	_____ **/ 10**
2026-10-29	0230650	chore: update clinician dashboard notes	
2026-10-28	14c27c3	feat: pass SHAP feature data to clinician dashboard	
2026-10-28	5319e5e	feat: pass ECG signal data to clinician dashboard	
2026-10-28	edf54e7	perf: optimize ECG waveform rendering	
2026-10-22	1ef4c3b	feat: improve clinical AI report UI	
2026-10-22	4d6ed2c	feat: improve ECG waveform visualization	
2026-10-22	20b1ac1	feat: improve SHAP explainability UI	
2026-10-22	4544187	feat: improve patient registration UI	
2026-10-15	6a4cf2c	chore(reports): organize weekly reports and automation	
2026-10-15	98be38a	test(backend): expand API and database validation coverage	
2026-10-15	13ce79a	feat(ml): update multimodal datasets and model artifacts	
2026-10-15	255cc12	feat(backend): improve auth data access and ECG APIs	
2026-10-15	ea9e6c4	fix(frontend): repair dashboard styling and role workflows	
2026-10-08	7cde138	chore(frontend): restore ESLint 9 configuration	
2026-10-08	9c89957	refactor(frontend): normalize dashboard sample data	
2026-10-08	6663762	feat(frontend): surface dashboard test-data connection status	
2026-10-08	6e8ca63	fix(frontend): keep dashboard accessible during API outages	
2026-10-08	e172a36	fix(frontend): route patient registration through API client	

Date	Hash	Commit Message	Mentor Marks (/10)
2026-10-08	cec9a79	feat(frontend): connect login and registration to JWT API	
2026-10-08	09cf1de	feat(frontend): protect dashboard routes with session auth	
2026-10-08	81e537a	feat(frontend): add API and auth session services	
2026-10-01	33f1ad7	feat: add AI clinical report component	
2026-10-01	1ec9e15	feat: add ECG analysis and AI clinical report	
2026-10-01	e688174	feat: add AI clinical report component	
2026-10-01	2fa8fe9	feat: add ECG analysis and AI clinical report	
2026-10-01	37dea73	feat: add SHAP explanation chart UI	
2026-10-01	1b43088	feat: enhance clinical dashboard overview	
2026-10-01	5a37f16	feat: connect dashboard with test clinical data	
2026-09-25	1f47868	feat: enhance sidebar and responsive clinical interface	
2026-09-24	1b9db19	feat: enhance clinician navigation bar	
2026-09-24	02b7add	feat: improve clinical dashboard and registration workflow	
2026-09-24	869c49a	feat: improve patient registration validation	
2026-09-24	e7a8763	feat: enhance clinician dashboard workflow	
2026-09-16	0c7660f	feat: add authentication routes and styling	
2026-09-16	4e64f2d	feat: add registration page	
2026-09-16	40adca7	feat: add login page	
2026-09-16	cd53ba4	feat: connect patient registration to database API	
2026-09-10	486c528	docs: remove outdated weekly progress report	
2026-09-10	300921b	chore: recreate frontend project	
2026-09-10	1f33b0a	chore: remove existing frontend	
2026-09-08	6d6af46	ci: add automated weekly Form-3 workflow	
2026-09-08	75e1582	ci: add automated Form-3 report generator	
2026-08-09	ef37a46	Initial CVD-XAI project with frontend	

Student: Jatin Khandelwal — 39 commit(s)

Date	Hash	Commit Message	Mentor Marks (/10)
2026-10-29	4deb27e	docs(db): add Sprint 3 live explainability pipeline specification and update work report	_____ **/ 10**
2026-10-29	c6fe0b8	test(api): add unit test coverage for patient explainability and global SHAP endpoints	
2026-10-29	0b91ff9	feat(db): add check_shap_pipeline audit tool to verify MongoDB XAI storage and indexing	
2026-10-29	49a9a29	feat(frontend): connect dashboard service to live MongoDB SHAP explainability endpoint	
2026-10-29	97a62e6	feat(api): add live SHAP explainability endpoint and granular attribution persistence in patient_api.py	
2026-10-22	785ac49	docs(db): add Sprint 3 SHAP storage specification and update comprehensive work report	
2026-10-22	14f242d	test(db): add unit test coverage for SHAP explanation storage and global feature ranking	
2026-10-22	a7b5a6a	feat(db): add seed_shap_explanations utility to populate clinical explainability cohorts	
2026-10-22	1bc2dca	feat(db): implement SHAP explainability persistence engine and global feature aggregation in database.py	
2026-10-22	58a6364	feat(db): extend PredictionResult schema with granular SHAP waterfall and multimodal explainability fields	
2026-10-15	198e4b7	docs(db): add Sprint 3 model training organization spec and update work portfolio	
2026-10-15	7f77419	test(db): add automated unit test suite for training data extraction and derivations	

Date	Hash	Commit Message	Mentor Marks (/10)
2026-10-15	f6b6e9e	feat(db): add verify_model_data_alignment tool to audit MongoDB and ML model compatibility	
2026-10-15	336d640	feat(db): add prepare_training_data pipeline matching Deepak's 16-feature vector specification	
2026-10-15	04fbbf9	feat(db): add training cohort extraction and multimodal pairing helpers in database.py	
2026-10-08	6348685	docs(db): update test metrics to 38 passing tests and document XAI prediction queries	
2026-10-08	aefe0e8	feat(db): add realistic patient cohort seeding for clinician dashboard testing	
2026-10-08	0d18733	test(db): add unit test coverage for patient lookup and prediction history	
2026-10-08	d0ea178	feat(db): add patient prediction history query helper for XAI outputs	
2026-10-08	a525c8b	docs(db): add comprehensive Form 2 sprint work portfolio and sync healthcheck index suite	
2026-10-06	cf5c970	test: add coverage for ECG history helper, fix stale collection list	
2026-10-06	e9a09bb	feat(db): add ECG history helper + missing compound index (groundwork for ECG endpoint)	
2026-10-06	361265b	test: add coverage for ECG metadata import mapping	
2026-10-06	ffc7a7b	feat(db): import real ECG metadata (34,905 records) instead of fake seed data	
2026-10-06	f1bbb0f	docs: verify ECGRecording schema compatibility with new CNN-BiLSTM model	
2026-10-06	343a751	fix(schema): correct ECGRecording lead_count default to match 12-lead model	
2026-09-25	a67a401	docs: document ClinicalRecord vs ClinicalPredictionInput schema gap	
2026-09-25	fb6b322	test(db): add mongomock-based tests for database.py query functions	
2026-09-22	044fa0c	docs: add MongoDB Atlas setup guide for new team members	
2026-09-22	414549c	docs: document prediction_results collection	
2026-09-22	63905bf	feat(db): add pending health-check script, updated for prediction_results	
2026-09-14	11037a0	test: add coverage for PredictionResult schema	
2026-09-14	2a57547	feat(db): add PredictionResult schema for SHAP/LIME storage (ahead of Sprint 3)	
2026-09-14	6269b96	perf(db): add compound index, patient history helper, and dataset import script	
2026-09-13	39d6a94	test: add coverage for categorical field validation	
2026-09-13	17c8eb3	fix(validation): restrict categorical clinical fields to valid enum values	
2026-09-12	fd7acb8	test(db): add unit tests and docs for MongoDB schema	
2026-09-10	c7105f5	feat(db): migrate database layer from PostgreSQL to MongoDB	
2026-09-10	e1ae1d5	feat(db): migrate database layer from PostgreSQL to MongoDB	

Student: Deepak Kumar Saini — 49 commit(s)

Date	Hash	Commit Message	Mentor Marks (/10)
2026-10-22	30b02a8	Add clinical model hyperparameter tuning notebooks	**_____ / 10**
2026-10-22	f8a1d8e	Evaluate stacking ensemble classification results	
2026-10-22	148baa0	Optimize multimodal fusion and ensemble models	
2026-10-22	d985e14	Train stacking ensemble for CVD prediction	
2026-10-22	a7d8a6c	Improve ECG and clinical model performance	
2026-10-15	cab238a	Add final multimodal fusion pipeline script	
2026-10-15	a7fa82c	Finalize multimodal fusion workflow	
2026-10-15	32a9731	Check fusion pipeline configuration	
2026-10-15	60245f0	Review multimodal feature alignment	
2026-10-15	1547599	Validate clinical and ECG model integration	

Date	Hash	Commit Message	Mentor Marks (/10)
2026-10-15	92312f4	Prepare final multimodal fusion pipeline	
2026-10-09	f779e01	database input data check rules changing	
2026-10-08	c27b40e	Ignore ECG preprocessing cache	
2026-10-08	e74ed0a	Add SHAP explainability results for multimodal model	
2026-10-08	9e7d639	Implement clinical and ECG multimodal fusion	
2026-10-08	b6a33e7	Add clinical feature extraction and SHAP explainability	
2026-10-08	934cd46	Retrain clinical DNN model	
2026-10-08	34fba0f	Retrain ECG CNN-BiLSTM model	
2026-10-02	a57c5d0	Ignore raw Ningbo ECG dataset	
2026-10-02	8d4ae07	Add ECG preprocessing and CNN-BiLSTM model	
2026-10-02	d59a661	Improve ECG model performance	
2026-10-02	dc0b44f	Train and evaluate ECG classification model	
2026-10-02	7a64dd0	Prepare ECG data for model training	
2026-10-02	f0c0ea5	Improve ECG signal preprocessing	
2026-10-02	31a8761	Analyze ECG dataset structure	
2026-09-25	eeb7368	Add ECG dataset analysis notebook	
2026-09-24	8fbf729	Review clinical and ECG modeling workflow	
2026-09-24	35d4809	Prepare ECG features for modeling	
2026-09-24	c931cac	Set up ECG data analysis	
2026-09-24	be83b23	Prepare ECG signal preprocessing	
2026-09-24	cf84c2f	Explore ECG dataset structure	
2026-09-24	8597d0b	Begin ECG dataset integration	
2026-09-24	df40531	Refine clinical DNN evaluation	
2026-09-24	c7a041b	Optimize clinical model performance	
2026-09-24	9dce9bd	Improve clinical model preprocessing	
2026-09-17	1442c20	Train and improve DNN model for CVD prediction and Save the best model files	
2026-09-17	1cbf1f4	Train and improve DNN model for CVD prediction	
2026-09-16	96c2b6f	Improve model performance with hyperparameter tuning and cross-validation	
2026-09-14	183a5d6	Ignore generated CatBoost logs	
2026-09-14	b13cfdc	Update data preprocessing notebooks	
2026-09-14	6a84ef6	Add cleaned clinical dataset	
2026-09-10	eed09b8	Add dataset preprocessing notebook and requirements	
2026-09-10	34c3b2f	Add dataset preprocessing notebook and requirements	
2026-09-07	855ba9c	Create models folder	
2026-09-07	f6c24d6	Create models folder	
2026-09-07	0155373	Delete Models file	
2026-09-07	d91ae3b	Delete Models file	
2026-08-10	de19136	Create Models	
2026-08-10	a4a0459	Create Models	

Student: kanishk-procoder — 4 commit(s)

Date	Hash	Commit Message	Mentor Marks (/10)
2026-09-18	41ffda0	tested the api and andded the api test file	_____ / 10
2026-09-17	67a5fe7	some changes made in the prediction api and main.py file	
2026-09-17	56f52e9	created clinical prediction api for dnn model	
2026-09-17	0114750	made changes in the existing code to solve the database connectivity issue and added .bat files to make running the app easy	

Student: goyalharshit029-create — 2 commit(s)

Date	Hash	Commit Message	Mentor Marks (/10)
2026-09-08	afad687	ci: add automated weekly Form-3 workflow	_____ / 10
2026-09-08	1860d42	ci: add automated Form-3 report generator	

Name: _____

Designation: Project Mentor

Signature: _____

Name: _____

Designation: Lab Coordinator

Signature: _____